

Tree traversals: Visiting all nodes in a tree (and practicing recursion)

5/2/25

Recursive methods

- Check for base case
- Otherwise, break the problem into 1 or more smaller problem instances. Solve them by calling the same method and use the results to solve the original problem.

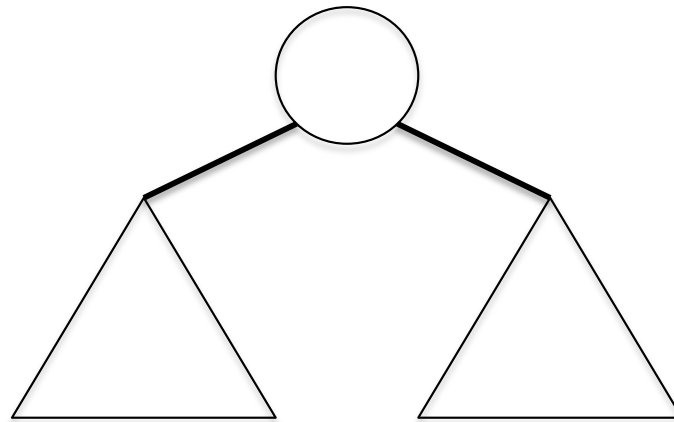
Example from text: countDown

```
public static void countDown(int n) {  
    //prints numbers from n down to 1  
    System.out.println(n);  
    if(n > 1)  
        countDown(n-1);  
}
```

Examples we've seen

- Mergesort
- Quicksort
- Arguably, binary search
- BST operations

Problem: Visit all nodes of a binary tree

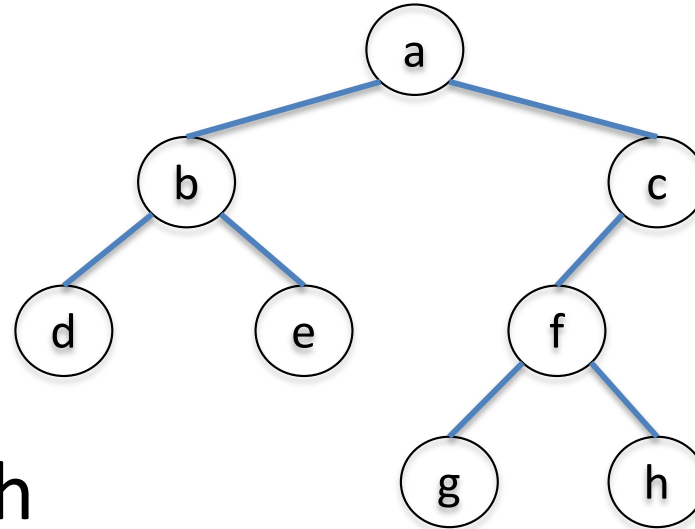


- Useful for printing the tree contents (for debugging, to store in a file, etc)
- Sometimes also need to visit all nodes to set flags, add the values, count something, ...

Tree traversals

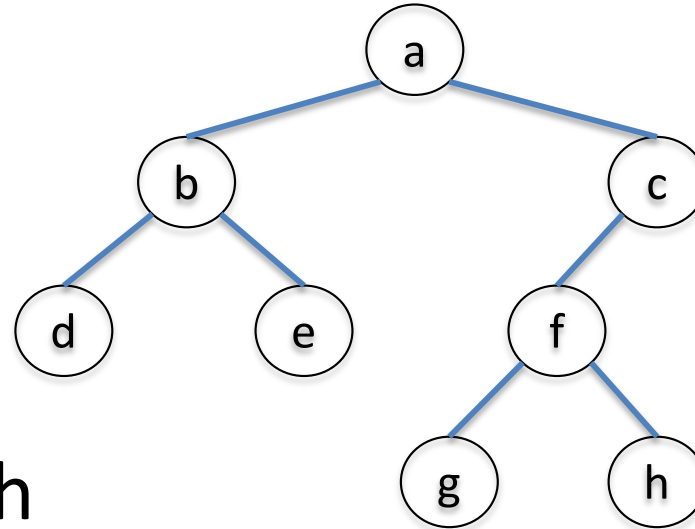
```
void traverse(Node n) {  
    if(n == null) return;  
    //preorder: print here  
    traverse(n.left);  
    //inorder: print here  
    traverse(n.right);  
    //postorder: print here  
}
```

Which of the following is a preorder traversal of the tree below?



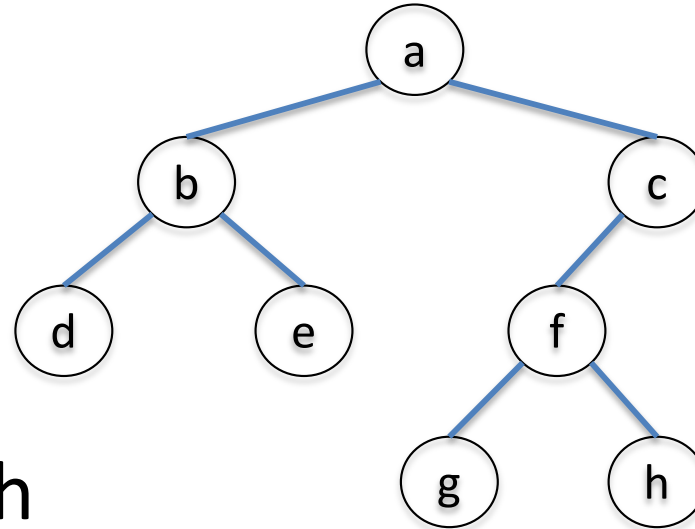
- A. a,b,c,d,e,f,g,h
- B. a,b,c,d,f,e,g,h
- C. a,b,d,e,c,f,g,h
- D. a,c,b,d,f,e,g,h
- E. None of the above

Which of the following is a preorder traversal of the tree below?



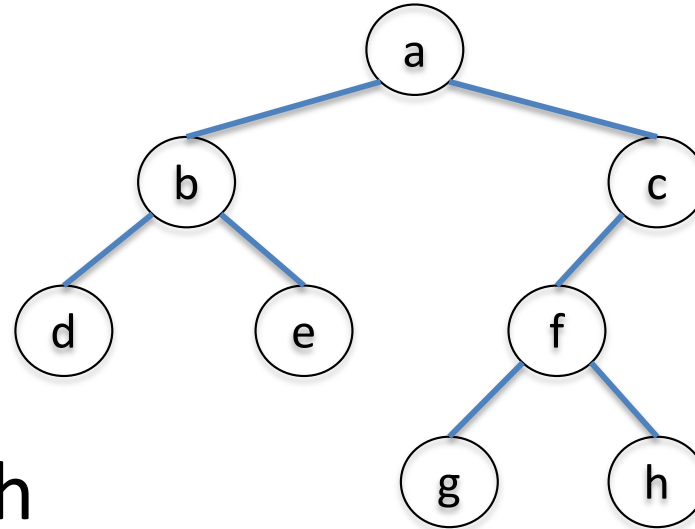
- A. a,b,c,d,e,f,g,h
- B. a,b,c,d,f,e,g,h
- C. a,b,d,e,c,f,g,h
- D. a,c,b,d,f,e,g,h
- E. None of the above

Which of the following is an inorder traversal of the tree below?



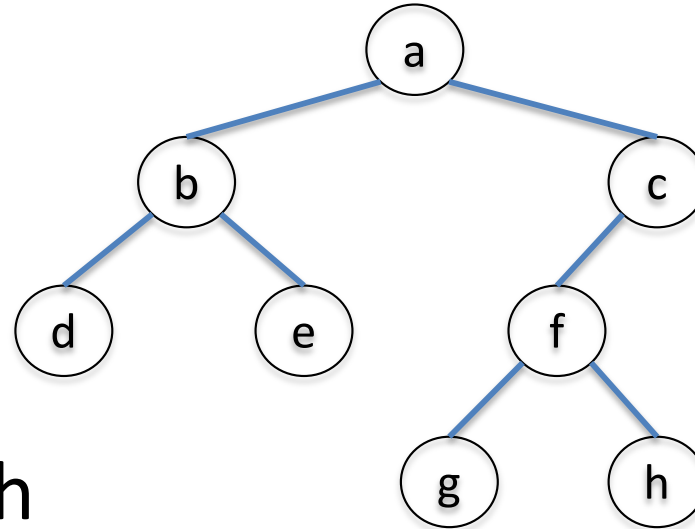
- A. a,b,c,d,e,f,g,h
- B. b,a,c,d,e,g,f,h
- C. d,b,e,a,g,f,h,c
- D. d,b,a,e,g,f,c,h
- E. None of the above

Which of the following is an inorder traversal of the tree below?



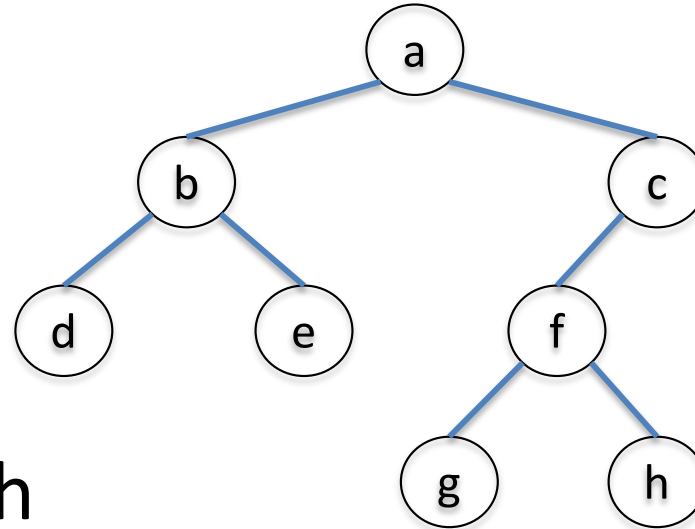
- A. a,b,c,d,e,f,g,h
- B. b,a,c,d,e,g,f,h
- C. d,b,e,a,g,f,h,c
- D. d,b,a,e,g,f,c,h
- E. None of the above

Which of the following is a postorder traversal of the tree below?



- A. b,d,e,a,f,c,g,h
- B. b,d,e,g,h,f,c,a
- C. d,e,b,a,g,h,f,c
- D. d,e,b,g,h,f,c,a
- E. None of the above

Which of the following is a postorder traversal of the tree below?



- A. b,d,e,a,f,c,g,h
- B. b,d,e,g,h,f,c,a
- C. d,e,b,a,g,h,f,c
- D. d,e,b,g,h,f,c,a
- E. None of the above

Draw two trees with preorder traversal a,b,c,d,e,f,g,h

Draw two trees with inorder
traversal a,b,c,d,e,f,g,h

Draw the unique tree with both

- preorder traversal:

a,b,c,d,e,f,g,h

- inorder traversal:

b,c,a,f,e,g,d,h

Alternate: Level-order traversal

- Visits the nodes one level at a time
(the root, its children, its grandchildren...)
- Implemented using a queue

Sample problem: Towers of Hanoi